

SQL Window Functions Intro

Project Report | ROW_NUMBER, RANK, DENSE_RANK and LAG on AdventureWorks 2022

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Tool: SQL (PostgreSQL / pgAdmin)

Dataset: AdventureWorks 2022 (denormalized) - 57,851 order lines, Jul 2017 to May 2020

Deliverables: 12 queries, outputs, notes

1. Objective and Overview

Is project ka goal tha window functions (ROW_NUMBER, RANK, DENSE_RANK, LAG) ko real business questions pe use karna: ranking, top-N per group, aur month-over-month / year-over-year trend. Window functions GROUP BY ki tarah rows ko collapse nahi karte - har row bani rehti hai aur saath me ek calculated value judti hai, jo 'OVER (PARTITION BY ... ORDER BY ...)' se control hoti hai.

2. Dataset and Preparation

Raw file pipe (|) delimited thi, money columns me '\$' aur ',' the, aur dates 'Friday, August 25, 2017' format me thi. prepare_data.py se file clean ki gayi: dates ko YYYY-MM-DD banaya, money columns numeric kiye, aur 12 zaruri columns rakhe. Clean data PostgreSQL ki table sales me import kiya gaya. Ek row = ek order line item; total_sales = quantity x unit_price.

Note: data Jul 2017 se May 2020 tak hai, isliye 2017 aur 2020 adhoore saal hain (year-over-year comparison me dhyan rakhna hai).

3. Queries, Outputs and Notes

Q1. ROW_NUMBER

Business question: Har order ke andar line items ko sales value (high to low) ke hisaab se number dena.

```
SELECT sales_order_number, product_name, total_sales,
       ROW_NUMBER() OVER (PARTITION BY sales_order_number ORDER BY total_sales DESC) AS line_rank
FROM sales
WHERE sales_order_number IN ('SO43666', 'SO43667', 'SO43676')
ORDER BY sales_order_number, line_rank;
```

Output (pehli 10 rows, total 15):

sales_order_number	product_name	total_sales	line_rank
SO43666	Road-150 Red, 56	2,146.96	1
SO43666	Road-450 Red, 44	874.79	2
SO43666	Road-650 Black, 44	838.92	3
SO43666	Road-650 Red, 52	419.46	4
SO43666	Road-650 Black, 60	419.46	5
SO43666	ML Road Frame - Red, 48	356.90	6
SO43667	Mountain-100 Silver, 44	2,039.99	1
SO43667	Mountain-100 Black, 48	2,024.99	2
SO43667	Mountain-100 Black, 38	2,024.99	3
SO43667	Mountain Bike Socks, L	17.10	4

Notes: SO43666 me 6 line items hain aur sabse badi line Road-150 Red, 56 (\$2,146.96) hai. SO43667 me do products ki value barabar hai (\$2,024.99), phir bhi ROW_NUMBER unhe alag numbers (2 aur 3) deta hai - ROW_NUMBER kabhi tie nahi dikhata.

Q2. ROW_NUMBER

Business question: Har reseller ka sabse recent order nikalna (top-1-per-group / dedup pattern).

```
WITH orders AS (
    SELECT sales_order_number, reseller_name, MAX(order_date) AS order_date,
           SUM(total_sales) AS order_value
    FROM sales
    GROUP BY sales_order_number, reseller_name
), numbered AS (
    SELECT *, ROW_NUMBER() OVER (PARTITION BY reseller_name ORDER BY order_date DESC, sales_order_number
                                DESC) AS rn
    FROM orders
)
SELECT reseller_name, sales_order_number, order_date, order_value
FROM numbered
WHERE rn = 1
ORDER BY order_value DESC
LIMIT 10;
```

Output (pehli 10 rows, total 10):

reseller_name	sales_order_number	order_date	order_value
Westside Plaza	SO67305	2020-04-16	116,729.89
Roadway Bicycle Supply	SO69531	2020-05-26	112,312.42
Perfect Toys	SO69535	2020-05-27	109,694.44
Thorough Parts and Repair Services	SO69508	2020-05-21	106,267.61
Rally Master Company Inc	SO67298	2020-04-14	102,725.70
Hardware Components	SO46067	2018-05-18	101,857.11
Great Bicycle Supply	SO46066	2018-05-18	100,378.71
Tenth Bike Store	SO50204	2019-05-04	97,596.87
Eastside Sporting Goods	SO49888	2019-04-27	96,605.72
Brakes and Gears	SO69471	2020-05-14	95,037.98

Notes: rn = 1 filter lagane se har reseller ki sirf ek row bachti hai. Westside Plaza ka latest order (Apr 2020) \$116,730 ka hai, jo sab resellers ke latest orders me sabse bada hai.

Q3. ROW_NUMBER

Business question: Har sales territory region ke top 3 products (total sales ke hisaab se).

```
WITH product_sales AS (
    SELECT sales_territory_region, product_name, SUM(total_sales) AS sales
    FROM sales
    GROUP BY sales_territory_region, product_name
), ranked AS (
    SELECT *, ROW_NUMBER() OVER (PARTITION BY sales_territory_region ORDER BY sales DESC) AS rn
    FROM product_sales
)
SELECT sales_territory_region, rn AS position, product_name, sales
FROM ranked
WHERE rn <= 3
ORDER BY sales_territory_region, rn;
```

Output (pehli 10 rows, total 30):

sales_territory_region	position	product_name	sales
Australia	1	Touring-1000 Yellow, 60	132,554.24
Australia	2	Touring-1000 Blue, 60	123,017.84
Australia	3	Touring-1000 Blue, 46	115,245.80
Canada	1	Mountain-200 Black, 38	609,027.43

sales_territory_region	position	product_name	sales
Canada	2	Mountain-200 Black, 42	482,783.63
Canada	3	Mountain-200 Silver, 38	415,356.89
Central	1	Mountain-200 Black, 42	232,690.44
Central	2	Mountain-200 Silver, 42	230,029.81
Central	3	Mountain-200 Black, 38	220,417.23
France	1	Mountain-200 Black, 38	254,173.00

Notes: Canada me Mountain-200 Black, 38 sabse upar hai (\$609K). Australia aur Germany me Touring-1000 variants lead karte hain - regions ke hisaab se product mix alag hai.

Q4. RANK

Business question: 17 salespeople ko total sales ke hisaab se rank karna.

```
SELECT salesperson_fullname,
       SUM(total_sales) AS total_sales,
       RANK() OVER (ORDER BY SUM(total_sales) DESC) AS sales_rank
FROM sales
GROUP BY salesperson_fullname
ORDER BY sales_rank;
```

Output (pehli 10 rows, total 17):

salesperson_fullname	total_sales	sales_rank
Linda Mitchell	10,158,634.52	1
Jillian Carson	9,755,992.01	2
Michael Blythe	8,952,751.48	3
Jae Pak	8,099,816.95	4
Tsvi Reiter	6,976,128.29	5
Shu Ito	6,283,193.33	6
José Saraiva	5,536,439.26	7
Ranjit Varkey Chudukatil	4,429,368.45	8
David Campbell	3,614,760.61	9
Garrett Vargas	3,486,102.49	10

Notes: Linda Mitchell #1 hai (\$10.16M), uske baad Jillian Carson (\$9.76M) aur Michael Blythe (\$8.95M). Kisi ki sales barabar nahi hai, isliye ranks 1 se 17 tak bina gap ke hain.

Q5. RANK + PARTITION BY

Business question: Har territory group ke andar top 3 salespeople.

```
WITH sp AS (
  SELECT sales_territory_group, salesperson_fullname, SUM(total_sales) AS total_sales
  FROM sales
  GROUP BY sales_territory_group, salesperson_fullname
), ranked AS (
  SELECT *, RANK() OVER (PARTITION BY sales_territory_group ORDER BY total_sales DESC) AS rnk
  FROM sp
)
SELECT sales_territory_group, rnk, salesperson_fullname, total_sales
FROM ranked
WHERE rnk <= 3
ORDER BY sales_territory_group, rnk;
```

Output (pehli 8 rows, total 8):

sales_territory_group	rnk	salesperson_fullname	total_sales
Europe	1	Ranjit Varkey Chudukatil	4,429,368.45
Europe	2	José Saraiva	3,440,481.64
Europe	3	Rachel Valdez	1,681,207.53
North America	1	Linda Mitchell	10,158,634.52
North America	2	Jillian Carson	9,755,992.01
North America	3	Michael Blythe	8,952,751.48
Pacific	1	Lynn Tsoflias	1,216,916.52
Pacific	2	Syed Abbas	174,108.33

Notes: Europe me Ranjit Varkey Chudukatil (\$4.43M), North America me Linda Mitchell, Pacific me Lynn Tsoflias lead karte hain. Pacific me sirf 2 salespeople hain, isliye sirf 2 rows aati hain.

Q6. ROW_NUMBER vs RANK vs DENSE_RANK

Business question: Tie wale products pe teeno functions ko side-by-side compare karna (units sold).

```
WITH units AS (
    SELECT product_name, SUM(quantity) AS units_sold
    FROM sales
    GROUP BY product_name
), ranked AS (
    SELECT product_name, units_sold,
           ROW_NUMBER() OVER (ORDER BY units_sold DESC) AS row_num,
           RANK() OVER (ORDER BY units_sold DESC) AS rnk,
           DENSE_RANK() OVER (ORDER BY units_sold DESC) AS dense_rnk
    FROM units
)
SELECT *
FROM ranked
WHERE rnk BETWEEN 100 AND 112
ORDER BY row_num;
```

Output (pehli 10 rows, total 13):

product_name	units_sold	row_num	rnk	dense_rnk
Road-250 Red, 48	647	100	100	100
Road-350-W Yellow, 42	642	101	101	101
Men's Bib-Shorts, L	633	102	102	102
Mountain-100 Black, 38	633	103	102	102
HL Mountain Frame - Black, 38	626	104	104	103
HL Road Pedal	623	105	105	104
Road-550-W Yellow, 44	623	106	105	104
ML Mountain Frame - Black, 44	619	107	107	105
Mountain-100 Black, 44	618	108	108	106
LL Mountain Front Wheel	617	109	109	107

Notes: Men's Bib-Shorts, L aur Mountain-100 Black, 38 dono ke 633 units hain. ROW_NUMBER: 102 aur 103. RANK: 102 aur 102, aur agla rank 104 (103 skip). DENSE_RANK: 102 aur 102, agla 103 (koi gap nahi).

Q7. DENSE_RANK + PARTITION BY

Business question: Har reseller country me top 2 resellers.

```
WITH rs AS (  
    SELECT reseller_country, reseller_name, SUM(total_sales) AS total_sales  
    FROM sales  
    GROUP BY reseller_country, reseller_name  
) , ranked AS (  
    SELECT *, DENSE_RANK() OVER (PARTITION BY reseller_country ORDER BY total_sales DESC) AS drnk  
    FROM rs  
)  
SELECT reseller_country, drnk, reseller_name, total_sales  
FROM ranked  
WHERE drnk <= 2  
ORDER BY reseller_country, drnk;
```

Output (pehli 10 rows, total 12):

reseller_country	drnk	reseller_name	total_sales
Australia	1	Nationwide Supply	166,967.27
Australia	2	Rich Department Store	154,664.79
Canada	1	Retail Mall	803,767.27
Canada	2	Vigorous Exercise Company	794,909.73
France	1	Metropolitan Equipment	648,107.78
France	2	Registered Cycle Store	582,454.82
Germany	1	Global Bike Retailers	343,435.41
Germany	2	Best Cycle Store	278,251.20
United Kingdom	1	Metropolitan Bicycle Supply	482,282.01
United Kingdom	2	Bulk Discount Store	400,253.07

Notes: United States me Brakes and Gears (\$882K) aur Canada me Retail Mall (\$804K) apne country ke #1 reseller hain. DENSE_RANK tab safe hai jab 'top N distinct levels' chahiye.

Q8. LAG

Business question: Monthly sales ke saath pichhle mahine ki sales ek hi row me laana.

```
WITH monthly AS (  
    SELECT DATE_TRUNC('month', order_date)::DATE AS month, SUM(total_sales) AS sales  
    FROM sales  
    GROUP BY 1  
)  
SELECT month, sales,  
       LAG(sales) OVER (ORDER BY month) AS prev_month_sales  
FROM monthly  
ORDER BY month;
```

Output (pehli 10 rows, total 35):

month	sales	prev_month_sales
2017-07-01	489,328.40	
2017-08-01	1,540,072.02	489,328.40
2017-09-01	1,166,332.45	1,540,072.02
2017-10-01	844,832.71	1,166,332.45
2017-11-01	2,325,754.91	844,832.71
2017-12-01	1,703,435.09	2,325,754.91
2018-01-01	713,229.54	1,703,435.09

month	sales	prev_month_sales
2018-02-01	1,900,794.46	713,229.54
2018-03-01	1,455,280.27	1,900,794.46
2018-04-01	883,010.98	1,455,280.27

Notes: Pehle mahine (Jul 2017) ki prev_month_sales NULL hai kyunki usse pehle koi row nahi. LAG se self-join ki zarurat khatam ho jaati hai.

Q9. LAG

Business question: Month-over-month change aur growth % nikalna.

```
WITH monthly AS (
    SELECT DATE_TRUNC('month', order_date)::DATE AS month, SUM(total_sales) AS sales
    FROM sales
    GROUP BY 1
)
SELECT month, sales,
       sales - LAG(sales) OVER (ORDER BY month) AS change_amt,
       ROUND(100.0 * (sales - LAG(sales) OVER (ORDER BY month))
             / LAG(sales) OVER (ORDER BY month), 2) AS mom_growth_pct
FROM monthly
ORDER BY month;
```

Output (pehli 10 rows, total 35):

month	sales	change_amt	mom_growth_pct
2017-07-01	489,328.40		
2017-08-01	1,540,072.02	1,050,743.62	214.73
2017-09-01	1,166,332.45	-373,739.57	-24.27
2017-10-01	844,832.71	-321,499.74	-27.57
2017-11-01	2,325,754.91	1,480,922.20	175.29
2017-12-01	1,703,435.09	-622,319.82	-26.76
2018-01-01	713,229.54	-990,205.55	-58.13
2018-02-01	1,900,794.46	1,187,564.92	166.51
2018-03-01	1,455,280.27	-445,514.19	-23.44
2018-04-01	883,010.98	-572,269.29	-39.32

Notes: Har saal Jan me sales girti hai (2018: -58.13%, 2019: -39.74%, 2020: -52.68%) aur Feb me recover hoti hai. Aug 2017 ka +214.73% chhote base (Jul 2017 sirf \$0.49M) ki wajah se hai.

Q10. LAG + PARTITION BY

Business question: Territory group ke hisaab se year-over-year growth.

```
WITH yearly AS (
    SELECT sales_territory_group, EXTRACT(YEAR FROM order_date)::INT AS yr, SUM(total_sales) AS sales
    FROM sales
    GROUP BY 1, 2
)
SELECT sales_territory_group, yr, sales,
       LAG(sales) OVER (PARTITION BY sales_territory_group ORDER BY yr) AS prev_year_sales,
       ROUND(100.0 * (sales - LAG(sales) OVER (PARTITION BY sales_territory_group ORDER BY yr))
             / LAG(sales) OVER (PARTITION BY sales_territory_group ORDER BY yr), 2) AS yoy_growth_pct
FROM yearly
ORDER BY sales_territory_group, yr;
```

Output (pehli 9 rows, total 9):

sales_territory_group	yr	sales	prev_year_sales	yoy_growth_pct
Europe	2018	1,705,150.47		
Europe	2019	5,720,500.58	1,705,150.47	235.48
Europe	2020	2,862,974.99	5,720,500.58	-49.95
North America	2017	8,069,755.58		
North America	2018	22,623,635.78	8,069,755.58	180.35
North America	2019	25,903,883.18	22,623,635.78	14.50
North America	2020	9,271,644.77	25,903,883.18	-64.21
Pacific	2019	875,621.82		
Pacific	2020	515,403.03	875,621.82	-41.14

Notes: North America 2019 me +14.50% badha; Europe +235.48% (chhote base se). 2017 (Jul-Dec) aur 2020 (Jan-May) adhoore saal hain, isliye 2020 ka negative growth asli decline nahi hai.

Q11. LAG

Business question: Ek reseller ke do consecutive orders ke beech ke din (repeat purchase gap).

```
WITH orders AS (
    SELECT sales_order_number, reseller_name, MAX(order_date) AS order_date, SUM(total_sales) AS
        order_value
    FROM sales
    GROUP BY sales_order_number, reseller_name
), gaps AS (
    SELECT reseller_name, sales_order_number, order_date, order_value,
        order_date - LAG(order_date) OVER (PARTITION BY reseller_name ORDER BY order_date,
            sales_order_number) AS days_since_prev_order
    FROM orders
)
SELECT reseller_name, COUNT(days_since_prev_order) AS repeat_orders,
    ROUND(AVG(days_since_prev_order), 1) AS avg_gap_days
FROM gaps
GROUP BY reseller_name
HAVING COUNT(days_since_prev_order) >= 8
ORDER BY avg_gap_days
LIMIT 10;
```

Output (pehli 10 rows, total 10):

reseller_name	repeat_orders	avg_gap_days
Sports Products Store	10	61.90
Friendly Bike Shop	14	65.40
Trailblazing Sports	10	89.90
Riders Company	10	90.00
Excellent Riding Supplies	10	90.10
Larger Cycle Shop	10	90.10
Remarkable Bike Store	9	90.20
Basic Sports Equipment	11	90.30
Primary Bike Distributors	11	90.30
Scratch-Resistant Finishes Company	8	90.30

Notes: Zyadatar resellers lagbhag har 90 din (quarterly) order karte hain. Sports Products Store sabse jaldi repeat karta hai (avg 61.9 din).

Q12. SUM() OVER + LAG

Business question: Year-to-date running total aur pichhle mahine se Up/Down comparison.

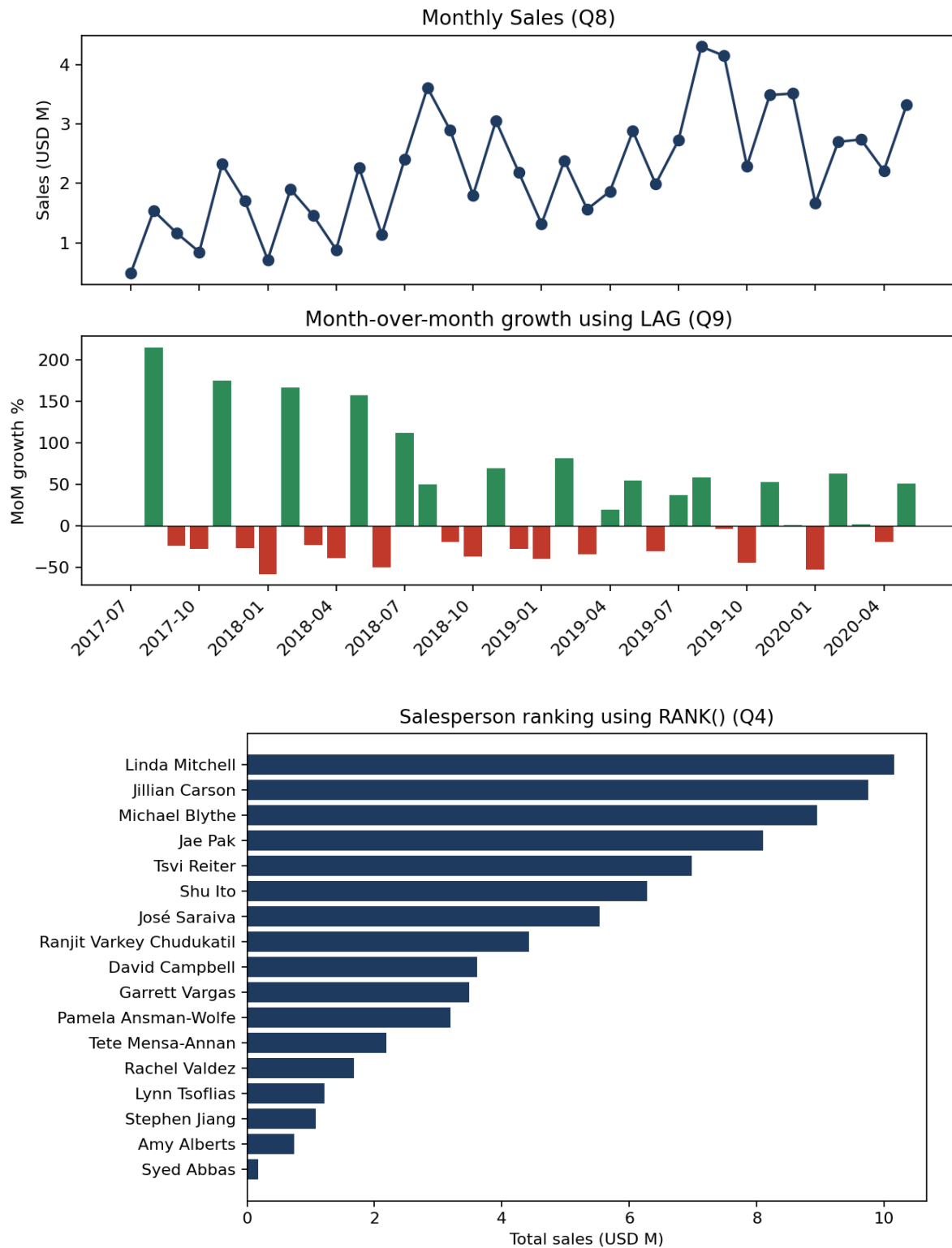
```
WITH monthly AS (  
    SELECT EXTRACT(YEAR FROM order_date)::INT AS yr,  
           DATE_TRUNC('month', order_date)::DATE AS month,  
           SUM(total_sales) AS sales  
    FROM sales  
    GROUP BY 1, 2  
)  
SELECT yr, month, sales,  
       SUM(sales) OVER (PARTITION BY yr ORDER BY month) AS ytd_sales,  
       CASE WHEN sales > LAG(sales) OVER (PARTITION BY yr ORDER BY month) THEN 'Up'  
            WHEN sales < LAG(sales) OVER (PARTITION BY yr ORDER BY month) THEN 'Down'  
            ELSE '-' END AS vs_prev_month  
FROM monthly  
ORDER BY month;
```

Output (pehli 10 rows, total 35):

yr	month	sales	ytd_sales	vs_prev_month
2017	2017-07-01	489,328.40	489,328.40	-
2017	2017-08-01	1,540,072.02	2,029,400.42	Up
2017	2017-09-01	1,166,332.45	3,195,732.87	Down
2017	2017-10-01	844,832.71	4,040,565.58	Down
2017	2017-11-01	2,325,754.91	6,366,320.49	Up
2017	2017-12-01	1,703,435.09	8,069,755.58	Down
2018	2018-01-01	713,229.54	713,229.54	-
2018	2018-02-01	1,900,794.46	2,614,024.00	Up
2018	2018-03-01	1,455,280.27	4,069,304.27	Down
2018	2018-04-01	883,010.98	4,952,315.25	Down

Notes: Running total se har saal ka cumulative dikhta hai: 2018 me \$24.33M, 2019 me \$32.50M tak pahunchi. PARTITION BY yr se total har saal reset hota hai.

4. Charts



5. ROW_NUMBER vs RANK vs DENSE_RANK

Function	Ties handle?	Gaps after tie?	Use case
ROW_NUMBER	Nahi - tie par bhi alag number	Nahi	Dedup, top-1 per group
RANK	Haan - same rank	Haan (1,2,2,4)	Olympic-style ranking
DENSE_RANK	Haan - same rank	Nahi (1,2,2,3)	Top-N distinct levels

6. When to use LAG

LAG(col, n) pichhli n-th row ki value laata hai (same partition ke andar, ORDER BY ke hisaab se). Use cases: month-over-month / year-over-year growth, do events ke beech ka time gap (repeat purchase), aur price ya status change detect karna. Pehli row ke liye pichhli row nahi hoti, isliye NULL aata hai (ya LAG(col, 1, default) use karo).

7. Key Insights

- Linda Mitchell, Jillian Carson aur Michael Blythe top 3 salespeople hain (total \$28.9M).
- Har saal January me sales girti hai aur February me recover hoti hai - seasonality.
- Resellers ka typical repeat-order gap lagbhag 90 din hai (quarterly buying).
- Top product region ke hisaab se badalta hai (Canada: Mountain-200, Australia/Germany: Touring-1000).

8. Conclusion and Learnings

Window functions se bina self-join ya subquery ke ranking aur trend analysis ek hi query me ho jaata hai. Sabse bada learning: ROW_NUMBER, RANK aur DENSE_RANK tie hone par alag behave karte hain, aur 'top N per group' ke liye ROW_NUMBER + CTE + WHERE rn <= N ek standard pattern hai.

Files: window_functions_queries.sql, prepare_data.py, outputs/*.csv, charts, README.md